



Effects of trees on their recruits in the southern Appalachians, USA

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ABSTRACT

The negative effect of conspecific trees on seedling recruitment in temperate forests has been well documented at the population level for several common species. In 2007, we estimated the survival of 2210 recently germinated seedlings of nine tree species transplanted near conspecific and heterospecific trees, a surrogate for describing distance-dependent mortality, as part of an experiment with landscape-level replication across eight mixed-deciduous forests in the southern Appalachian Mountains of North Carolina. Forest composition was variable but they had a number of woody species in common. Prior to establishing the field experiment, we used a forest inventory database for the region to classify the recruitment patterns of tree species and formulate predictions for species. The field experiment, conducted as a drought was progressing, revealed that four of the nine species planted had variable survival around conspecifics compared with heterospecifics suggesting variation in distance-dependent mortality. *Acer saccharum* and *Tsuga canadensis* both had greater mortality near conspecifics than heterospecifics, while *Fagus grandifolia* and *Prunus serotina* showed the opposite pattern. Species classified as having greater recruitment around conspecifics, according to the forest inventory data, suffered greater overall levels of mortality in our field experiment. Possibly because of the progressing drought, none of the four species predicted to be most affected by distance-dependent sources of mortality based on the forest inventory data exhibited the predicted patterns of survival near conspecific vs. heterospecific trees in the field experiment. Furthermore, two of the four species (*A. saccharum* and *T. canadensis*) classified as being least affected by conspecific trees actually had greater survival near heterospecifics than conspecifics. Although we identified effects of canopy tree type in four of the nine comparisons, negative effects of conspecific trees were observed for only two (*A. saccharum* and *T. canadensis*) of nine species and mostly contradicted predictions based on patterns from forest inventory data. The inconsistency between patterns from the forest inventory data and from experiments indicates that there may be localized, complex interactions that make generalizations about neighbor effects on tree seedling survival difficult.

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1. Introduction

Several studies indicate that temperate trees often have a negative effect on conspecific recruitment (e.g. Woods, 1979; Streng et al., 1989; Jones and Sharitz, 1998; Packer and Clay, 2000; Hille Ris Lambers et al., 2002). A recent study found that the relative abundance of tropical trees was positively correlated with their soil feedback effect suggesting that rareness is caused by negative distance-dependent effects caused by soil-borne pathogens (Mangan et al., 2010). The implied significance of these forms of negative frequency-dependent effects (i.e. non-competitive distance- or density-dependent effects) is that they provide additional explanations for how forest communities are structured, species co-exist, and forest diversity is maintained relative to niche partitioning theory which emphasizes intra- and inter-specific

competition. However, a recent meta-analysis testing the commonality of this distance-dependent effect was inconclusive after examination of data from 40 tropical and temperate studies (Hyatt et al., 2003). For temperate systems comprising five studies, the meta-analysis results tended to show the opposite pattern – mortality increased with increasing distance from the parent tree (Hyatt et al., 2003). The general conclusion from this meta-analysis was that distance-dependent effects of canopy trees are important for numerous populations but are not universally important across species (Hyatt et al., 2003). However, their meta-analysis emphasized tropical studies more than temperate, included studies often only partly related to forest dynamics, and did not include literature published after 1998.

Future studies attempting to quantify distant-dependent effects of canopy trees would benefit by screening species *a priori* and not assuming that all species will exhibit patterns of distance-dependent mortality. To do this, we established a quantitative metric for ranking the recruitment patterns of tree species, which

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characterized recruitment of seedlings and saplings around conspecific trees versus recruitment in areas without conspecific trees. These data were used to calculate an *Inhibition* index for species common in the southern Appalachian Mountains. Species with most of their recruits occurring in areas without conspecific trees were considered more likely to be affected by non-competitive distance- or density-dependent sources of mortality (i.e. Janzen-Connell Hypothesis [Janzen, 1970; Connell, 1971]). Specifically, we tested whether our *a priori* index of *Inhibition* corresponds with variation in susceptibility to herbivores and disease in the field to gain an improved understanding of the importance of negative distance-dependent effects on forest community assembly. In this study, tree species from the two extremes, described as inhibitors (i.e. rarely recruit in plots with conspecific trees) vs. facilitators (i.e. commonly recruit in plots with conspecific trees), were selected and used in a regional experiment with landscape-level replication across eight sites to test whether seedling mortality of planted seedlings was greater around conspecific than heterospecific trees. We predicted that species classified as inhibitors would exhibit greater mortality around conspecifics than heterospecifics compared to species classified as facilitators. Concordance of pattern from the forest inventory database and the field experiments would reveal the utility of using forest inventory data to inform the design of field experiments. In addition, we compared survival of seedlings classified as either facilitators or inhibitors and whether classification of heterospecific trees affects seedling survival. This was done to determine overall variation in the seedling survival of the two groupings of seedlings and whether generalizable differences in recruitment around different classes of heterospecific trees exist.

2. Materials and methods

2.1. Recruitment patterns

We used the US Forest Service *Forest Inventory and Analysis Database* (FIADB) to document recruitment patterns of tree species and to characterize the variation among temperate trees in the southern Appalachians region of the USA (<http://fia.fs.fed.us/>). The FIADB consists of systematic randomly located plots across the United States, a larger subplot (7.31 m in radius) where tree species over 12.7 cm in D.B.H. are recorded and a nested microplot (2.07 m in radius) where seedlings and saplings less than 12.7 cm D.B.H. and taller than 0.305 m are recorded. Specifically, we queried the database to estimate the probability that recruits of a species occur in microplots nested in subplots with trees of the same species vs. recruits in microplots nested in subplots without trees of the same species. Plots were excluded if they were in tree plantations or only partially forested. We queried the FIADB to select subplot and microplot data from portions of the central Appalachian broadleaf forest-coniferous forest-meadow province and the eastern broadleaf forest province (Bailey, 2001) located in eastern Tennessee and western North Carolina to create a dataset that was comparable to the intended study region. We omitted tree species from our database if they were present in fewer than 50 subplots.

We calculated inhibition index, $I = \ln(\text{total \# of subplots with adult trees and recruits of species X} / \text{total \# of subplots with recruits of species X but no conspecific adult trees})$.

Using randomly simulated data, this equation produces values that are symmetrically distributed around zero. After determining each species' *Inhibition* index, species were ranked by their index values and grouped based on their position along this continuum. An *Inhibition* index value would be greater than zero when recruits on average occur in areas that coincide with conspecific trees than

areas without conspecific trees. Negative values indicate a pre-dominance of recruitment in areas without conspecific trees. Trees with relatively large inhibition values are referred to as facilitator species and vice versa for species classified as inhibitor species. This index is comparable to distance-dependent recruitment patterns often used as evidence for the Janzen-Connell Hypothesis (e.g. Reinhart and Clay, 2009). This description of recruitment was used to guide selection of species used in the seedling survival experiment (see below).

2.2. Seedling survival experiment

Across all sites, we experimentally tested effects on nine species with four classified as inhibitors (i.e. low probability of recruiting near conspecifics [Fig. 1]), four as facilitators (i.e. highest probabilities of recruiting near conspecifics), and one intermediate. Per site, we experimentally tested nearly an equal number of facilitator and inhibitor species (see Table 1). In October 2006, we located eight sites in the Nantahala National Forest of the southern Appalachian Mountains in western North Carolina containing two or more facilitator (mean = 2.50 per site) and inhibitor species (mean = 2.13) to balance our sampling design and provide landscape-level replication. Descriptions of the sites and sampling design are shown in Table 1. We did not measure tree density per area but a separate analysis of the FIADB revealed that *Inhibition* index values were positively correlated with mean tree density such that species with the largest mean densities per sample area tended to have *Inhibition* index values in the upper range limits (DJ unpublished data). We selected relatively flat sites which were ca. 1 ha in size and generally surrounded by sloping topography. During an initial survey of the sites, we selected the facilitator and inhibitor species that were most common across all sites (Fig. 1). Although two to three representative facilitator and inhibitor tree species were identified per site (average of 4.63 total species per site), the variability among sites in the region caused the identities of the focal species to vary by site (Table 1).

We purchased seed for the nine tree species from a commercial seed supplier (Sheffield's Seed Co., Inc. Locke, NY, USA). To have all species germinate at approximately the same time, each received varying cold stratification, warm-moist stratification, or boiling treatments. In anticipation of having difficulty having all species germinate simultaneously, we initiated multiple batches of seeds separated by 2–4 weeks relative to an anticipated duration of pre-treatment to help ensure a sufficient supply of seedlings of each species. Ultimately, the availability of seedlings did limit the number of sites that could be planted with some species. Transplants were recently germinated seedlings and were at the cotyledon-stage of development. Plots were established around conspecific and heterospecific trees and planted with a total of 2210 seedlings during May 19–25, 2007. Two plots were established 1.5–2 m away from each focal tree. Each plot was 0.9×0.5 m with eight 10×10 cm cells separated from each other by 10 cm. Five seedlings of a species were planted into a cell whose location was randomly selected and mapped. A cell contained plantings of species found at that site (gray cells shown in Table 1) and contained either conspecific or heterospecific species. Only one cell per plot was selected for each heterospecific species ($5 \text{ seedlings} \times \text{heterospecific species}^{-1} \times 1 \text{ grid cell}^{-1} \times 2 \text{ plots}^{-1} \times \text{focal tree}^{-1}$) and two cells per plot were used to plant five seedlings of the conspecific species ($5 \text{ seedlings} \times 2 \text{ grid cell}^{-1} \times 2 \text{ plots}^{-1} \times \text{focal tree}^{-1}$). By mapping the location of seedlings and concentrating where the seedlings were planted, we were able to simulate natural establishment and avoid confusion that might result from natural recruitment. Plots were watered once immediately after planting. One of the two plots was treated with a granular fungicide (Subdue GR, Syngenta Crop Protection, Greensboro, North Carolina, USA) to

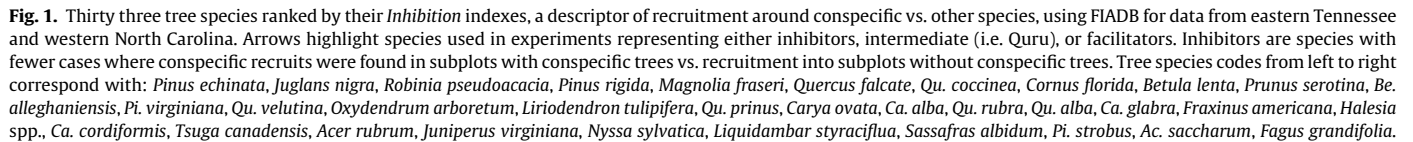


Table 1
Mesic deciduous forest sites used in the seedling transplant experiment. Sites contained species classified as inhibitor, facilitator, and intermediate (i.e. *Quercus rubra*) classes based on forest inventory data. Sites were in one of two regions either the Nantahala National Forest (NNF) west of Otto, NC or around Highlands, NC. The number of seedlings planted per species and site is shown in the gray cells. 20 seedlings were planted near a conspecific and 10 seedlings were planted around each heterospecific species shaded in gray.

Region	Site	Lat/Long	Elev. (m)	Inhibitor				Facilitator				Intermediate
				<i>Betula lenta</i>	<i>Prunus serotina</i>	<i>Quercus coccinea</i>	<i>Robinia pseudoacacia</i>	<i>Acer rubrum</i>	<i>Acer saccharum</i>	<i>Fagus grandifolia</i>	<i>Tsuga canadensis</i>	
NNF, Otto	1	N35 02.10 W83 28.82	1332	70	70		70		70	70	70	
NNF, Otto	2	N35 01.89 W83 30.29	1134	60			60	60		60	60	
NNF, Otto	3	N35 04.13 W83 31.30	1073		70		70	70			70	70
NNF, Otto	4	N35 01.87 W83 28.90	1348		60		60	60			60	
NNF, Highlands	5	N35 01.53 W83 10.46	958			60	60				60	60
NNF, Highlands	6	N35 02.31 W83 08.47	933	50			50				50	
NNF, Highlands	7	N35 06.23 W83 12.49	1321	50	50			50			50	
NNF, Highlands	8	N35 04.45 W83 15.59	1031	50			50	50			50	

limit the effects of soil-borne pathogens following the methods of a previous study (Reinhart and Clay, 2009). Seedling survival and attack, damage caused by disease and herbivory combined, were quantified in each cell on June 17–18, 2007. At the conclusion of the experiment, we did not observe any visual signs of soil-borne disease (e.g. damping-off) regardless of fungicide plot treatment. Since there was also no statistical evidence that the fungicide treatment had any effect on seedling survival (data not shown), it was excluded from any further comparisons. Lack of damping-off in our seedlings may be attributable to a regional drought (moderate to severe drought, according to <http://droughtmonitor.unl.edu/>) that progressed with the timing of the experiment and likely suppressed soil-borne diseases which are typically associated with moist conditions (e.g. Augspurger, 1990; Agrios, 1997).

We fit the data on seedling survival with a generalized linear mixed model (GLMM) using Proc GLIMMIX in SAS version 9.2 (SAS Institute 2007). A binomial distribution and a logit function were used since data represent the number of dead/attacked seedlings out of five seedlings per cell. The first analysis was performed for each species separately and used focal tree type (conspecific vs. heterospecific), site, and region (see Table 1) as variables. Tree type (inhibitor vs. facilitator) was treated as a fixed effect. Random effects included site(region) for species that had replication across regions and sites. Species with limited site replication had different random effects (see Table 1); *Acer saccharum* only had site as a random effect and *Fagus grandifolia*, *Quercus coccinea*, and *Q. rubra* had no random effects modeled. Additional analyses were performed using survival and attack data when seedlings were only planted around heterospecific trees and included tree classification (facilitator vs. inhibitor) and seedling classification as variables. For this analysis, tree and seedling classifications were treated as fixed effects.

3. Results

3.1. Recruitment patterns

After limiting the FIADB query to species that were present in more than 50 subplots, descriptions of *Inhibition* indexes were calculated for 33 tree species having a mean of 271 subplots occupied by that species (Fig. 1). We ranked species according to their inhibition index from most inhibitive (<-2 , e.g. *Pinus rigida*, *Pinus echinata*, *Juglans cinerea*, *Robinia pseudoacacia*, etc.) to most facilitative (e.g. *F. grandifolia*, *A. saccharum*, *Tsuga canadensis*, *Acer rubrum*, etc.). Species classified as inhibitor species have comparatively limited recruitment in subplots with conspecific trees relative to species ranked to the right (referred to as facilitator species). The variation in recruitment patterns among species informed the design of the field experiment.

3.2. Seedling survival experiment

The type of focal tree (conspecific vs. heterospecific) adjacent to where seedlings were planted affected the survival of four out of nine species but did not correspond with predictions based on recruitment patterns interpreted from forest inventory data. Seedlings of *A. saccharum* and *T. canadensis* both suffered greater mortality near conspecifics than heterospecifics (GLMM, $F_{1,36} = 11.7$, $P = 0.0016$ and $F_{1,85} = 3.9$, $P = 0.052$, respectively). In contrast, *F. grandifolia* and *Prunus serotina* suffered greater mortality near heterospecifics than conspecifics (GLMM, $F_{1,24} = 8.6$, $P = 0.0072$ and $F_{1,45} = 15.3$, $P = 0.0003$, respectively) (Fig. 2). The survival of other species was not affected by tree type ($P \geq 0.19$). Variation in mortality with neighboring tree type did not conform to

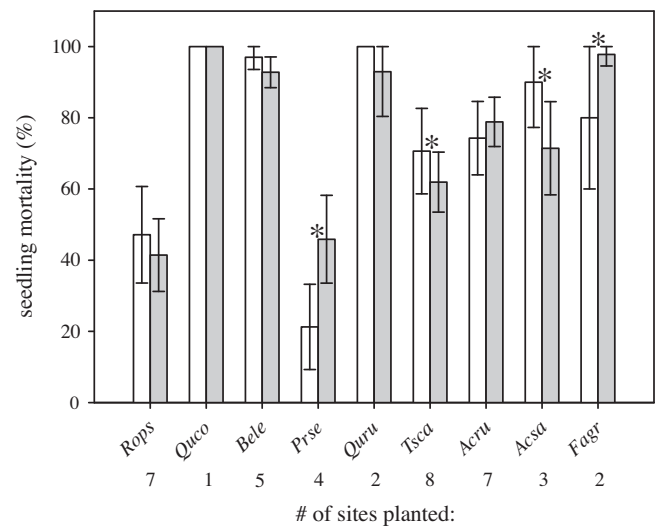


Fig. 2. Effect of planting location (conspecific vs. heterospecific) on seedling mortality and 95% CI. Seedlings planted near conspecific trees are depicted with white shaded bars and near heterospecific trees have gray shaded bars. Species are ranked along the x-axis according to their *Inhibition* index ranking (see Fig. 1). Species codes defined in Fig. 1. Number of sites planted is from Table 1. * $P < 0.05$.

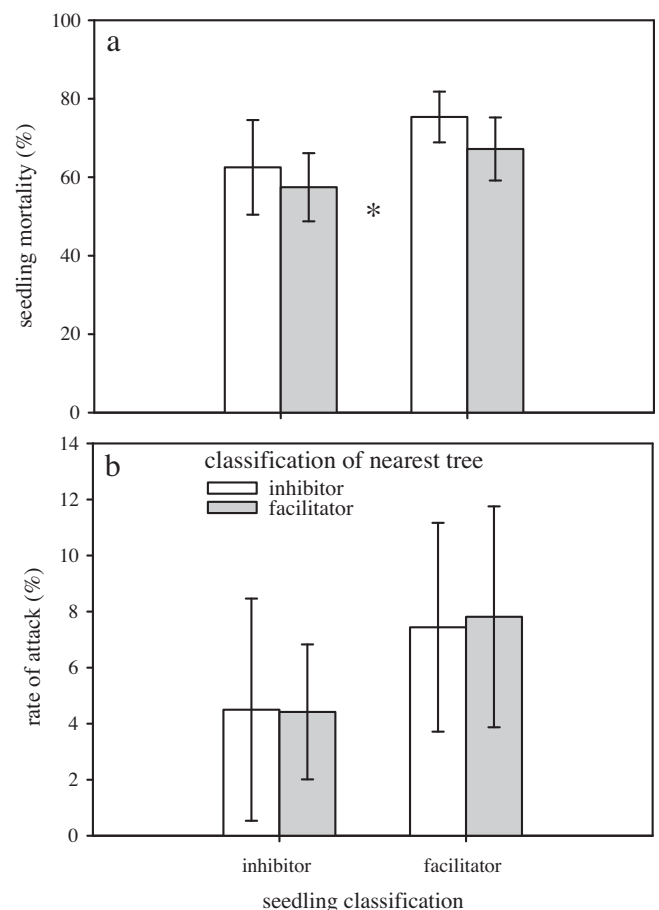


Fig. 3. Effect of tree and seedling classifications on the (a) survival and (b) level of attack of seedlings and 95% CI. Tree and seedling species were classified as either inhibitors or facilitators based on Fig. 1. Data are only for plantings next to heterospecific trees classified as inhibitors (white shaded bars) and facilitators (gray shaded bars). * $P < 0.05$.

predictions based on patterns from the forest inventory data. Our *a priori* prediction was that species classified as inhibitors would suffer proportionally greater mortality near conspecific than heterospecific trees. None of the four species classified as inhibitor species exhibited this response. Also contrary to our predictions, mortality was greater near conspecific than heterospecific trees for two (*A. saccharum* and *T. canadensis*) of the four species classified as facilitators.

We also determined whether variation in recruitment patterns among species in the field experiment corresponded to how seedling species were classified using the forest inventory data. Specifically, if seedlings of inhibitor species generally suffered more mortality than facilitator species and if survival rates differed when planted near heterospecific trees classified as inhibitors vs. facilitators. We compared the survival and attack rates associated with facilitator and inhibitor seedling groups around heterospecific trees of both groups. Across all facilitator and inhibitor species, there was no effect of heterospecific tree classification on seedling survival (GLMM, $F_{1,265} = 0.22$, $P = 0.64$) (Fig. 3a) indicating that mortality did not vary based on the grouping of the focal tree. Seedling species classified as facilitators experienced greater overall seedling mortality than seedling species classified as inhibitors (GLMM, $F_{1,265} = 19.9$, $P < 0.0001$) and no interaction between heterospecific tree class and seedling class was detected ($F_{1,265} = 0.37$, $P = 0.54$). Rate of attack did not vary by heterospecific tree class (GLMM, $F_{1,265} = 0.06$, $P = 0.81$), seedling class ($F_{1,265} = 2.90$, $P = 0.090$), and there was no interactive effect of tree and seedling classifications ($F_{1,265} = 0.01$, $P = 0.92$) (Fig. 3b).

4. Discussion

Previous studies have shown considerable variation among tree species in whether distance-dependent sources of mortality affect seedling populations (Hyatt et al., 2003). Two relevant studies conducted in southeastern US floodplains (Streng et al., 1989; Jones and Sharitz, 1998) and another in Appalachian forests near our current study (Hille Ris Lambers et al., 2002) report that many woody species exhibited distance-dependent sources of mortality. Unlike the present study, these studies describe patterns in mortality of naturally-occurring seedlings. To overcome the inconsistent finding of distance-dependent effects across species and studies, we used forest inventory data including data throughout the study region to classify species based on their recruitment patterns. Grouping species based on their patterns of seedling establishment provides a measure for estimating which species are affected by distance-dependent sources of mortality. Others have shown a linkage between tree demography patterns and measures of biological interactions (Mangan et al., 2010) that may cause distance-dependent mortality. In our field experiment, we observed that seedling survival varied around conspecific vs. heterospecific trees in four of nine species comparisons. We predicted that four species (*R. pseudoacacia*, *Q. coccinea*, *Betula lenta*, and *P. serotina*) would be most strongly affected by distance-dependent sources of mortality. However, three of these four species were unaffected by the type of tree (conspecific vs. heterospecific) they were planted near, and the other (*P. serotina*) experienced greater mortality near heterospecific than conspecific trees. Results for the four facilitator species (*T. canadensis*, *A. rubrum*, *A. saccharum*, and *F. grandifolia*) also differed from our predictions. Specifically, *T. canadensis* and *A. saccharum* species exhibited greater mortality near conspecific than heterospecific trees while *F. grandifolia* exhibited the opposite pattern.

Classifying species as either facilitators or inhibitors, based on forest inventory data, did not improve our ability to identify distance-dependent sources of mortality. One possibility for this

discrepancy is the forest inventory database is quantifying “seedlings” in microplots that are at least 0.305 m tall. These are surviving seedlings that may tell us little about mortality of recently emerged seedlings. The cumulative effect of processes occurring in nature that affect the forest inventory data may not necessarily be characterized by a short term field experiment. Another is that distance-dependent processes may not be accurately approximated by comparing survival around conspecific vs. heterospecific trees. A series of studies exploring the distance- and density-dependent nature of soil-borne disease of *P. serotina* have revealed spatial variation in patterns of soil-borne disease (Packer and Clay, 2000; Reinhart et al., 2005; Reinhart and Clay, 2009). Distant-dependent sources of mortality have been reported to vary by species (Hyatt et al., 2003) and may reflect in part spatial and temporal variation not captured by experiments with limited replication over space and time. For example, none of the seedlings in our experiment revealed symptoms of damping-off disease caused by soil-borne pathogens while seedlings of *P. serotina* and *T. canadensis* in controlled soil inocula experiments using soils from many of the same sites did exhibit damping-off disease symptoms (K.O.R. personal observation). Additionally, the result for *P. serotina* survival in the field corresponded with differences in biomass production observed in a separate lab experiment testing the effect of soil inocula from conspecific vs. heterospecific trees of the same region (K.O.R. unpublished data). This was not the case for *T. canadensis* (K.O.R. unpublished data). Others have argued that effects of soil near conspecific vs. heterospecific on plant performance misses important variation occurring at the species-level because data are artificially aggregated by heterospecific classification (McCarthy-Neumann and Kobe, 2010). Our field experiment may have been further complicated by the regional drought (moderate to severe drought, according to <http://droughtmonitor.unl.edu/>) that progressed during the field experiment. This likely increased the prevalence of seedling mortality caused by water stress while decreased the amount caused by soil-borne disease (e.g. Augspurger, 1990; Agrios, 1997). This climatic event may have effectively obscured biotic effects that might be detected during a season with more typical precipitation. Other possible factors could include variation in mast years (seedling density) and the relatively short duration of the field experiment.

A novel result from our study was that seedlings of species classified as facilitators experienced greater overall mortality than species described as inhibitors. This is interesting because our *Inhibition* classes approximate traditional successional classifications. For example, many of the species classified as facilitators (e.g. *A. saccharum*, *F. grandifolia*, and *T. canadensis*) are typically classified as late-successional species (e.g. Baker, 1949). Other research results indicated that plant resistance to foliar herbivory increased with successional status (Cates and Orians, 1975). In addition, a study comparing the susceptibility of 21 tropical tree species to soil-borne disease revealed that shade-tolerance was positively correlated with resistance (McCarthy-Neumann and Kobe, 2008). Other studies also show that susceptibility to soil-borne pathogens may affect species' successional ranking (Augspurger, 1984; O'Hanlon-Manners and Kotanen, 2004; Reinhart et al., 2010; but see O'Hanlon-Manners and Kotanen, 2006). In contrast, results from our study, quantifying mostly abiotic effects of drought than effects of soil-borne pathogens, suggest that facilitator species (i.e. late-successional species) experience greater levels of seedling mortality than inhibitor species (Fig. 3). Variation among these studies and the roles of biotic vs. abiotic factors support the notion that various trade-offs exist and affect successional classifications. Physiological trade-offs are well documented that show variation between early- and late-successional species and how growth and survival is favored in high vs. low light (e.g. Kaelke et al., 2001). Biological trade-offs may also exist

relating to susceptibility to pathogens and herbivores (Cates and Orians, 1975; O'Hanlon-Manners and Kotanen, 2004; McCarthy-Neumann and Kobe, 2008; Reinhart et al., 2010). The timing of our field experiment likely occurred during a period when seedlings were subjected to atypical drought conditions that resulted in greater mortality due to water stress than the more typical abiotic stresses associated with low light availability. We interpret that the observed differences in survival among groups of species is likely a product of their ability to grow rapidly after transplanting, while conditions were most favorable, and to be sufficiently established (i.e. new root growth) to endure the developing drought. Early initial seedling growth has been positively correlated with reserve seed dry mass until that resource is depleted (e.g. Saverimuttu and Westoby, 1996; Bloor and Grubb, 2003). Though seed size explains some of the differences here, the species with the greatest survival (e.g. *P. serotina* and *R. pseudoacacia*) had seed sizes that were above average while other species with much greater mortality (e.g. *A. saccharum*, *F. grandifolia*, and *Q. coccinea*) had similar to larger seed sizes (unpublished data).

Theory predicts and empirical evidence often supports the idea that negative effects of soil-borne pathogens should accumulate over time, reducing local host recruitment and maintaining species coexistence (e.g. Packer and Clay, 2000; Reinhart et al., 2005). Several studies on *P. serotina* in other portions of its native range indicate greater pathogenic activity of soil associated with conspecifics than heterospecifics (Packer and Clay, 2000; Reinhart et al., 2005) and decreasing pathogenic effects with increasing distance from *P. serotina* trees (Reinhart and Clay, 2009). However, distance-dependent seedling mortality was not always observed and variation among sites, experiments (i.e. laboratory vs. field experiments), and even among individual trees have been reported (Packer and Clay, 2000; Reinhart et al., 2005; Reinhart and Clay, 2009). There is some evidence suggesting that in comparison to other tree species, *P. serotina* is more susceptible to plant disease than many other tree species (O'Hanlon-Manners and Kotanen, 2006; Reinhart et al., 2010).

Our results indicate that the most species diverse temperate forests in the USA (Currie and Paquin, 1987) do not appear to conform to predictions regarding distance-dependent mortality (e.g. Janzen-Connell and related hypotheses) at least when these forests are experiencing moderate to severe drought conditions. Our study, which used broad regional sampling and was not designed to test for many species-level effects per site, identified several responses that contradicted our predictions. Other sources of competitive (e.g. nutrient and water depletion) and non-competitive distance or density-dependent sources of mortality (e.g. small mammals, slugs, etc.) may be structuring these temperate forests and driving the recruitment patterns described by our *Inhibition* index (Fig. 1). However, results from our field experiment suggest that early seral/inhibitor species tended to experience less mortality than later seral/facilitator species (Fig. 3a). Overall, positive or negative feedback processes occurring either above- or belowground appeared relatively scarce in the field and a separate soil inocula experiment (K.O.R. unpublished data) which may be a consequence of their spatial and temporal heterogeneity within and among sites, and possibly the drought conditions that existed during the field experiment. *P. serotina* has a record of being highly susceptible to belowground attack but transplanted seedlings had relatively high rates of survival compared to other species (Fig. 2) suggesting that trade-offs might cause certain species to be differentially affected by abiotic and biotic factors. If a component of regional species diversity is also variability in interactions with biota above- and belowground then it may be extremely challenging to generalize species interactions (i.e. distant-dependent mortality) across a region. In deciduous forests of the lower Midwest (US),

we have documented considerable spatial variation in soil-borne pathogens and their effects on *P. serotina* (Reinhart and Clay, 2009). Results from our study and others (O'Hanlon-Manners and Kotanen, 2006) suggest that complex biotic and abiotic interactions are structuring temperate forests. However, if spatial variability is common in systems, it may limit the prospects of identifying biotic interactions that drive recruitment patterns of multiple species across broad regions.

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